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Studying as Self-Regulated Learning

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Asked what they are doing as they engage in school- or university-related activities, students might say, "I'm working on my project," "I'm doing homework," or, commonly, "I'm studying." Charting the boundaries of studying is a fuzzy task. Colloquially, the range of activities it encompasses is nearly synonymous with all the activities that probabilistically lead to "a relatively permanent change in cognitive structure."¹ However, several features may distinguish studying from the subsuming category of learning activities. In particular, studying:

1. Rarely includes direct or frequent intervention by a teacher.
2. Is often a solo activity, although peer mediation is also common.
3. Often originates with a general goal set by a teacher that the student subsequently interprets at the studying session's outset and refines in a recursive way as studying unfolds.
4. Quite often involves searching in and synthesizing information from multiple sources, for example; a text book, notes taken in a class or borrowed from a friend, a volume of an encyclopedia, a video or TV show, or online databases (e.g., PsychLit).
5. Quite often occurs in settings where the student can engineer the studying environment to satisfy personal preferences (e.g., studying with or without the radio turned on, automatic calls that check email, or stimulants supplied by tea or coffee).

¹We use the phrase *cognitive structure* rather than the term *knowledge* because we include some topics that are not typically included in the category labeled by the term *knowledge*. In particular, these topics include propositions about motivation (e.g., an efficacy expectation about competence to enact a study tactic under current conditions) and about affect (e.g., positive feelings about applying a study tactic).

6. Almost always produces observable traces (Winne, 1982) of cognitive processing in forms such as notes in a notebook or in the margins of a textbook's pages, outlines, summaries, self-generated questions, diagrams, records of attempts to solve problems, and especially highlighted (or underlined) text.

Studying has been a topic of research for nearly a century. The dominant form of research has been to compare unknown "normal" studying activities to a specific study tactic—underlining, outlining, making notes as a spatial display of nodes and labeled links (concept maps or webs), generating questions, and so forth—that is described in terms of observable traces or what students were trained to use. Less common is research in which students are observed or trained to use a set of study tactics that are coordinated into a strategy, such as the 6-step method of previewing, questioning, reading, reflecting, reciting, and reviewing (PQ4R; Thomas & Robinson, 1972). Reynolds and Shirey (1988) judged that both kinds of research on studying skills, tactics, and strategies overshadow work investigating more basic and generative issues about how to model cognitive processing that underlies studying. Hence, too little is known about cognitive processes that are the proximal causes of whatever effects may be observed when students enact specific study tactics or strategies. We concur.

PROSPECTUS FOR THE CHAPTER

The six features just enumerated that distinguish studying from learning in general describe circumstances that essentially compel students to engage in complex bundles of goal-directed cognitive and motivational processes that "get studying done." We view these bundles as instances of metacognitively powered self-regulated learning (Winne, 1995a, 1995b, 1996).

As a first step toward examining studying through metacognitive lenses, we present a general typology that delineates facets of academic tasks in general, including studying tasks. Then, we use this typology to characterize four distinguishable but recursively linked stages of studying: task definition, goal setting and planning, enacting study tactics and strategies, and metacognitively adapting studying. Next, we develop connections between our typology for studying and models of metacognitive monitoring, metacognitive control, and self-regulated learning (Butler & Winne, 1995; Winne, 1995a, 1995b). With this backdrop, we then survey select research that highlights metacognitive activities in each of the four stages of studying. Finally, we summarize our model of studying and offer suggestions for next steps in research on studying as a complex, self-regulated learning event.

THE COPES MODEL OF STUDY TASKS

Studying tasks vary enormously. We use a 5-facet typology as a schema to characterize instances of studying in terms of common dimensions: conditions, operations, products, evaluations (of products), and standards (for products). The typology, presented in Table 12.1, is referred to by the first-letter acronym COPES because it identifies facets of tasks that a student copes with in studying and learning (see Winne, 1997; Winne & Marx, 1989). In Table 12.1, we offer four instances of COPES facets that correspond to each of four stages of studying described next.

Four Stages of Studying

We theorize that a complete model of studying has four basic stages: *task definition*, *goal setting and planning*, *enactment*, and *adaptation* (see also Simpson & Nist, 1984). In the task definition stage, the student generates a perception about what the studying task is, and what constraints and resources are in place. Based on this perception or definition of the task, at Stage 2, the student selects or generates idiosyncratic goal(s) and constructs a plan for addressing that study task. In the enactment stage, the plan of study tactics that was created in Stage 2 is carried out. In Stage 4, based on the overall experience of this studying event at the first three stages, the student makes changes to cognitive structure that will affect future studying tasks. This forward planning changes knowledge, skills, beliefs, dispositions, and motivational factors that the student predicts will play important roles in future studying tasks.

All four stages of studying have the same general cognitive architecture (see Fig. 12.1). A collage of environmental factors and cognitive information constitutes the *conditions* within which cognitive activities occur. Single cognitive *operations* and coordinated sets of them (tactics, strategies) create internal *products* by transforming conditions (Winne, 1989). For example, previously inert information in long-term memory may be stimulated and made available to working memory, information that is not organized may be assembled into a structured form, information represented in one medium (e.g., words) may be translated into another medium (e.g., algebraic symbols, an image), an inference or an attribution may be generated, and so forth. Operations also can create external *performance*, behaviors the student enacts that others can observe. Our four stages of studying are distinguished by the product created at each stage.

TABLE 12.1
Facets in the COPES Typology of Tasks

Facet	Definition	Stage 1: Task Definition	Stage 2: Goals & Plan	Stage 3: Enactment	Stage 4: Adaptation
Conditions	conditions that affect how the task will be engaged, including conditional knowledge (IFs in IF-THEN rules that filter which operations are enacted)	• interest • goal orientation • learning styles • time constraints • available resources • knowledge of tactics • task knowledge • subject matter expertise	• Stage 1 conditions + • Stage 1 products	• Stage 1 & 2 conditions + • Stage 2 products	• Stage 1-3 conditions + • Stage 1-3 products
Operations	cognitive processes, tactics, and strategies the student engages to address the task	• searching • monitoring (including tactics such as self-questioning, charting similarities and differences) • assembling (including tactics such as elaborating, integrating) • rehearsing (including tactics such as copying notes, rereading) • translating (including tactics such as making diagrams, 1st-letter mnemonics)			
Products	information created by operations	• Perception of what the task is, its COPES	• standards for the task as in Stage 1 or reframed in light of individual differences (e.g., learning style) • plan for coordinating study tactics • standards for enacting tactics (e.g., speed)	• traces of study tactics (e.g., margin notes, answers to questions, concept map) • new/reorganized subject matter knowledge	Updates to: • COPES that distinguish tasks • motivation, beliefs • conditional knowledge and study tactics • standards for judging qualities of tactics
Evaluations	feedback about products, either generated internally by the student or provided by external source(s)	Judgments about: • task understanding • COPES of the task Comparisons with other sources of information	Judgments about: • complexity/difficulty • incentive of goal • ability to carry out plan • effort required	Judgments about: • learning (calibration) • utility of tactics • efficacy • attributions	Judgments about: • "distance" between prior version and adaptation • predicted effects of adaptation Other evaluations must await future tasks
Standards	criteria against which products are monitored	• grading criteria • past performance	• Stage 1 standards & products • effort/utility thresholds • motivation orientation	• Stage 2 standards and products + object-level standards & meta-level standards	• Stage 1-3 standards and products

We adopt the common assumption that studying, like other forms of behavior, is a goal directed activity (Winne, 1995a). In our model of studying, goals are represented as a multivariate profile of standards. Standards characterize ideal, optimal, or satisficing states. For example, a student may hold standards to study a chapter (a) in 30 minutes, (b) so that a judgment of comprehension about the whole chapter is high, while (c) allowing that judgments of learning for specific bits of information are low (e.g., see Schraw, 1994).

We theorize that products take two forms. Cognitively, products are represented as a multivariate profile of attributes. For example, at the stage of defining the studying task, a student with a performance motivational orientation might perceive an assignment to study a chapter as a task that involves reading (a) without deep processing while (b) marking (highlighting) definitions and principles for future review. We hypothesize that, if students were asked to describe what they perceived or understood about informational products they create within a stage of studying, they would name such attributes and report levels for them. Externally, products can be manifested as behavior or performance.

When students monitor a product's profile of attributes against the profile of standards, cognitive evaluations are generated that mark whether an attribute is on target, how much off target an attribute is, and whether there is a discrepancy between items listed as standards and attributes of a product. External evaluations also can be created by performance when, for instance, a statistics package for the social sciences (SPSS) program has an error or a friend complains that a summary is off target. If the student acts on evaluations, this is control by which elements in the collage of cognitive conditions may be altered; standards may be adjusted, added, or abandoned; and, operations of new kinds may be carried out.

Our 4-stage model, like models of self-regulated learning (Butler & Winne, 1995; Nelson & Narens, 1994; Winne, 1995a, 1995b, 1996), is a recursive, weakly sequenced system. In recursive systems, products of earlier stages (or steps within a stage) update conditions on which operations work during the next cycle of activity. For example, the product of Stage 1, a definition of the studying task, updates cognitive conditions that establish the information and resources used in Stage 2, generating goals and plans. Although studying generally proceeds from Stage 1 to 2 to 3 to 4, stages are weakly sequenced and do not necessarily unfold in order. Some studying tasks may be so familiar that defining the

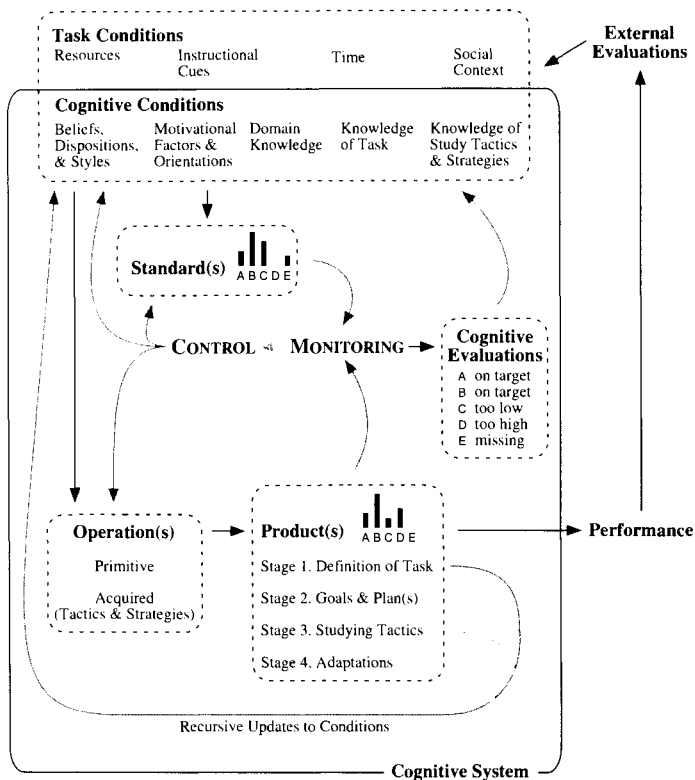


FIG. 12.1. The COPES model of metacognitive monitoring and control in four stages of studying.

task (Stage 1) is virtually skipped. Students who practice means–ends analysis will oscillate frequently between generating goals and plans (Stage 2) and enacting study tactics (Stage 3). Making adaptations can occur at practically any point.

Stage 1: Task Definition. In this stage, the student develops a perception about features of the task, charting the task as a “study space.” As we noted earlier, most study tasks are indeterminate with respect to one or several COPES facets. Thus, the student must make inferences about the nature of these facets. An important element in the product of this stage is the student’s perception of a goal for the task. A goal is a set of standards by which the task can be judged. The student’s perceived goal may be the same goal that the teacher or the text set. Or, it may be a goal that the student frames. Grade school students’ questions about whether the as-

signed summary of a science chapter will be handed in or whether they can collaborate on book reports are evidence of inferences about standards by which the task will be judged. In the next stage of studying, the student may include these standards in conditional knowledge that sets new goals, for example, to create neat work or to avoid behavior that will result in points deducted for “cheating.”

Latitude for students to develop perceptions about study tasks also characterizes interventions in experimental research. When students study a text, typical instructions might be: “Read the material and, if you have time, review. Don’t try to memorize every word, but be prepared for a test in 20 minutes.” Although students are explicitly instructed not to use tactics that create verbatim recall of the entire text (e.g., cumulative rehearsal), they may interpret the instructions as allowing main ideas to be rehearsed. They also are free to infer, for example, whether the text will be available during recall (a condition; probably not), whether they are allowed to underline or make marginal notes (modest sets of cognitive operations that comprise study tactics; if they have a pencil in a pocket, the inference is a 50–50 guess), and what kind of test will be administered (a standard; for veterans in subject pools, perhaps inferred to be free recall rather than multiple choice). Individual differences between students and intraindividual differences within a student over time practically ensure that there will be variance in perceptions about what a “given” study task is.

Stage 2: Goal Setting and Planning. In Stage 2 of studying, students may re-frame the goal(s) that were products of Stage 1 if the student’s personal standards differ from those that were first perceived for the task. For example, students with a performance motivational orientation that views tasks as just jobs to complete may judge that the goal they understood their teacher set is at too high a level or requires too much effort. Therefore, they adjust or alter standards for summarizing the science chapter to levels where “just getting by” is adequate. In light of this re-framed goal, the student now builds a plan to approach it. This student will probably plan simplistic tactics, such as paraphrasing headings and monitoring surface features of typography to insure that every bold phrase and every scientist’s name (standards) is reproduced in the finished product.

A student participating in the fictitious experiment we introduced in Stage 1 may set a goal to preserve time for deeply processing (a standard) sections of the text that are novel or complex (conditions). The plan for deep processing is to select that material (operation) and use a concept

mapping tactic (operation) that creates detailed views of the organization and elaborations (standards) described in the text. An enabling plan supports the student's first intention. It involves developing a general representation of the text's structure (product) using a tactic that can be applied quickly (a standard). That tactic is reading (operation) headings plus each paragraph's first and last sentences (task conditions).

Stage 3: Enacting Study Tactics and Strategies. Stage 3 of studying is enacting study tactics and strategies that the student has planned in Stage 2. Study tactics, such as making summaries, using first-letter mnemonics, underlining, and reviewing by re-reading headings are composites of operations by which the student addresses whatever task(s) the student perceived to be focal. As these operations are applied, evaluations (feedback) are generated internally. For instance, the grade school student may falter in paraphrasing the textbook's outline. Perhaps the student lacks vocabulary for expressing that content or has only shallow understanding about concepts in the subject matter domain. Monitoring this deficiency, the student may generate an attribution (evaluation): The task may be judged too difficult; or, if the student holds an entity belief about ability (condition), the student may attribute the deficiency to low ability.

For the university student participating in the experiment we introduced earlier, an unexpected announcement that "Only one quarter of the study period has lapsed" updates conditions. The student may monitor that, because deep processing is now in progress at the middle region of the text, too little time was spent developing a representation of the text's macrostructure (evaluation). There is spare time to add to the initial plan additional study tactics that consolidate the macrostructure and elaborate it with some details.

In this stage of studying, monitoring the products of study tactics may result in dynamically sensitive changes to the plan for studying. Our university student activated new study tactics not previously part of the plan generated in Stage 2 to take advantage of the situation as it had unfolded. In general, these kinds of instantaneous adaptations to the changing landscape of the task serve to repair gaps or errors in the student's evolving domain knowledge. An alternative adaptation in Stage 3 is that, if the student judges that there are no study tactics available to achieve goals, the student may quit the task.

Stage 4: Metacognitively Adapting Studying. In Stage 4, the student inspects products created in preceding stages, monitoring them relative

to metaknowledge that characterizes standards for those events. Monitoring at this stage is not about fine-tuning the process within just Stage 1 or 2 or 3. Rather, decisions made in Stage 4 take one of two forms. In one form, they address how activities coordinate across several stages of studying. Acting on these decisions results in large scale adjustments to the student's understandings about the task, goals, plans, and tactical engagement. The second form of adaptive decision making reaches beyond the boundaries of the present studying task to change the conditions for studying in the future. Acting on these latter decisions results in relatively permanent changes to cognitive conditions that will be in force for future studying tasks. A change like this might reshape motivational orientation from task-based to mastery-based. Or, the student may lower personal standards for success. Alternatively, by adding constraints to the conditional knowledge that triggers a tactic, the student may specialize the tactic's use for the future.

When students make deliberate changes to studying "on the fly," at any of the four stages of studying, they are engaging in metacognition. We now turn our focus to this topic.

METACOGNITION IN STUDYING

We propose that each stage of studying should be considered in terms of two main factors. The first factor is the student's expertise in the subject matter domain. Students' subject matter expertise reduces the need to allocate substantial cognitive resources to studying because domain knowledge itself is sufficient to make progress or achieve success (Schneider, 1993). For example, Körkel (1987; cited in Schneider, 1993) found that students' metacognitive knowledge about generic cognitive tactics for studying text combined linearly with the extent of domain knowledge in accounting for amount recalled. Domain novices (in soccer) with more metacognitive knowledge about generic studying tactics recalled more than novices with less metacognitive knowledge. The same benefit of metacognitive knowledge was observed among domain experts. Experts with less knowledge about cognitive tactics recalled less about what they had studied than experts with more knowledge of study tactics. Thus, when experts study material in their domain, although metacognition may be less frequent because it is less necessary (McKoon & Ratcliff, 1992; see also Wyatt, Pressley, El-Dinary, Stein, Evans, & Brown, 1993), metacognition can still benefit experts' studying.

The second factor to be considered in modeling studying is the degree to which the student is metacognitively active. Models of metacognition (Nelson & Narens, 1990, 1994) distinguish two “levels” of information—the object-level and the meta-level—and two operations—monitoring and control. In our 4-stage model of studying, *object-level information* corresponds to the products of cognitive operations that are created in each stage of studying: a definition of the task, goals and plans, tactics and strategies, and adaptations. *Meta-level information* in our model has two elements. Metastandards describe object-level conditions, operations, products, and standards. They constitute the student’s model of studying and provide the basis for metacognitively monitoring studying. The second element of meta-level information consists of cognitive operations that change object-level elements. This is metacognitive control.

Metacognitive Monitoring

Metacognitive monitoring is a cognitive operation that has two inputs and creates one product. One input is an object-level multivariate profile of attributes for a product of studying. The second input is a meta-level multivariate profile of attributes that constitutes a standard for that product. The product that metacognitive monitoring creates is a third multivariate profile of attributes, one that records discrepancies between the object-level information and the meta-level information. In our COPES typology, this cognitively generated product is an evaluation that serves as feedback.

For example, suppose a student is in Stage 3 of studying, enacting a study tactic. The tactic is mapping the correspondence between features of a principle that has been presented in a text and an example that the student is now studying (e.g., Walczyk & Hall, 1989). This is a monitoring tactic. It compares attributes that describe two forms of domain information, the principle and the example. The product this monitoring generates is a list of matches and non-matches. This is non-metacognitive monitoring. The standards that monitoring uses are neither a meta-level model of a principle nor a description of how to apply the study tactic.

However, suppose the student has a meta-level model that describes general standards for using study tactics. For a student who holds the epistemological view (cognitive condition) that learning should be quick and effortless (Schommer, 1994), three meta-level standards for studying would be: (a) time spent should be minimal, (b) effort (ease of learning judgments, Nelson & Narens, 1990; depth of search; see Winne, 1997) required to apply a study tactic should be minimal, and (c) the judgment of learning (Nelson & Narens, 1990) that follows using a study tactic should

be high. Monitoring other attributes that describe studying, such as actual time taken, effort spent, and strength of the judgment of learning contrasts this profile of meta-level standards to attributes that describe the product created by using the tactic. This is metacognitive monitoring. The product it creates is a profile of differences that describes how closely the study tactic as enacted matches meta-level standards for using study tactics in general.

Metacognitive Control and Self-Regulated Studying

Metacognitive control changes the task space at the object-level. The discrepancies created by metacognitive monitoring can be modeled as IFs in an IF-THEN (condition-action) rule. The THEN component of these rules is a cognitive operation (or a set of operations, a production system), such as a study tactic, that changes object-level information in the task space.

Suppose the student with the epistemological stance that learning should be quick and effortless monitors that, although time and effort are within tolerance of the meta-level standards, the judgment of learning is weak relative to the meta-level standard. This profile of attributes may trigger a 2-step production system:

1. IF time and effort spent are on target and
 IF judgment of learning is below standard,
 THEN attribute the negative difference to high task difficulty.
2. IF task difficulty is high,
 THEN quit the task.

This is self-regulated studying, task engagement in which students adapt² their approach to studying to reduce discrepancies revealed by metacognitive monitoring. Our example illustrates a form of self-regulation that is almost surely academically counterproductive (Winne, 1995a) because, based on idiosyncratically rational criteria (Anderson, 1991), it leads the student to quit the task for the “wrong” reasons. In parallel to negative effects of subject matter misconceptions (Pintrich, Marx, & Boyle, 1993) on learning in general, meta-level misconceptions (e.g., the double-edged sword; see Covington, 1992) can also interfere with studying.

Forms of Metacognitive Control. We suggest a distinction between

²On rational grounds (Anderson, 1991; see also Winne, 1995a), adaptation can include (a) maintaining forms of cognition that are presently being applied rather than changing them, and (b) quitting the task.

two forms of metacognitive control that exercises self-regulation, toggling and editing. With toggling, the student turns studying tactics or other cognitive operations on and off. Toggling immediately changes the task space by shaping the mix of operations carried out in a stage of studying.

With the second form of metacognitive control, editing, the student does more than just toggle operations on and off. In editing, changes are made to object-level cognitive conditions and study tactics. If changes are extended to meta-level standards for metacognitive monitoring, the student engages in mindful abstraction (Salomon & Perkins, 1989). Following Rumelhart and Norman (1978), we suggest that three kinds of changes can be made by editing:

1. **Accreting:** adding or subtracting (specializing or generalizing) (a) propositions in conditional knowledge that trigger study tactics, (b) component operations that enact study tactics, and (c) propositions that constitute standards for metacognitive monitoring.
2. **Tuning:** adjusting weights and thresholds that govern how propositions in conditional knowledge articulate in metacognitive monitoring; or, changing the sequence for performing operations in a study tactic.
3. **Restructuring:** substantially reconstituting conditional knowledge and significantly changing the mix of operations that make up studying tactics, including inventing new components.

Although toggling and all three of kinds of editing can be topics of explicit deliberation, very extended practice plus evaluations can make them automatic, pushing features of their use beneath reportable levels of attention (McKoon & Ratcliff, 1992). When this state is achieved, previously separate metacognitive activities have become composed as one smoothly running unit, and the student studies on self-regulated "auto-pilot." We note explicitly that the student's self-regulation may be judged by others' standards as deficient. Like other cognition, metacognitive aspects of studying are rationally adaptive (Anderson, 1991) in relation to criteria the individual adopts rather than criteria that others may hold "for" a student.

Summary

We model studying as an activity that generally follows a 4-stage sequence. First, the student forms a perception of what the task is. Goals and plans are then created, followed by enacting study tactics that the student predicts can achieve those goals. As operations are performed in each of these stages, evaluations are generated that the student may metacognitively monitor. Metacognitive monitoring also can occur at the end of a study session.

When a student metacognitively examines studying, metacognitive control can be exercised to toggle study tactics on and off, or editing may be done to adapt the conditions, operations, or standards in cognitive structure that describe studying. Toggling and editing are forms of self-regulated studying, but not all forms of self-regulation are productive (Winne, 1995a, 1997).

METACOGNITION IN STUDYING

A complete view of metacognitive engagement in studying at all four stages of studying encompasses several large classes of variables, as identified in Fig. 12.1. If the number and scope of these variables were not enough to make models of studying inherently complicated, our modeling of studying as a recursive event will. Although such models need not be chaotic, they can quickly explode in complexity.

Conditions in the external environment, including opportunities to receive external evaluations (Kluger & DeNisi, 1996), have been widely investigated in descriptive studies (e.g., questionnaire studies) and experiments. Conditions in the cognitive system, including feedback generated internally (see Butler & Winne, 1995) and variables that we loosely categorized as capacity or capability, are individual differences in cognitive structure. These also shape or, in the case of study tactics, are engines of studying and metacognition during studying.

Students' Definitions of Studying Tasks

A few studies investigate what students perceive about studying tasks. That researchers and students can differ in their respective perceptions is illustrated in a study by Córdón and Day (1996). They observed that students do use reading comprehension tactics while taking standardized readings tests, a possibility that some researchers had ruled out. Little research documents correspondences between how students study at home, in study halls, or in their dorms versus how they report or have been observed to study in research settings. Because conditions within each stage of studying can vary widely as a function of differences in studying environments, it would be prudent to generalize cautiously from the research we report.

Some individual differences that color perceptions of studying tasks may be global and quite unresponsive to task conditions. Schommer (1994) reviewed research on learner's systems of epistemological "beliefs about the nature of knowledge and learning ... [that] are likely to affect reasoning,

learning, and decision making" (p. 293). Her model describes five basic dimensions of such systems: source (handed down from authority vs. personally reasoned out), certainty (fixed vs. subject to revision), organization of knowledge (simple and compartmentalized vs. interwoven), control (predetermined vs. self-guided), and speed of learning (p. 301). These beliefs establish cognitive conditions that may lead students to adopt perceptions of tasks that pervade studying.

In one study (Schommer, 1990), undergraduates studied controversial texts that were missing a concluding paragraph on the resolution of the controversy, and then wrote a concluding paragraph. Qualities of their conclusions were predicted by epistemological beliefs. Undergraduates with stronger beliefs in quick learning tended to oversimplify the controversy. Those holding stronger beliefs in the certainty of knowledge were less likely to acknowledge that issues they had studied were indeterminate.

In another study (Schommer, Crouse, & Rhodes, 1992), undergraduates studied a text about statistics and were told to prepare for a multiple-choice test that contained items requiring both recognition and reasoning. After reading, but before the test, students estimated confidence in their understanding. Students' belief in simple-compartmentalized knowledge correlated negatively with learning but positively with overconfidence (relative to actual levels of performance). Students who held stronger beliefs in quick learning were more likely to report using shallow cognitive tactics for studying. A path analysis showed both direct and indirect effects on learning of the epistemological belief in simple knowledge.

Other studies suggest that not all perceptions of studying are inflexible. For example, Van Meter, Yokio, and Pressley (1994) interviewed college students about taking notes in lectures and how the students used notes to study subsequently. We report only a few illustrative findings. In general, these students perceived that forms of notetaking must adapt to task conditions, such as variations in lecturers' delivery speed and organization, and to cognitive conditions, such as their domain knowledge. Knowledge of notetaking tasks also was found. Notes record particular types of information, for example, material that is redundant with the text or that the professor stresses, but notes do not include information introduced by guest lecturers or in films. The students also were aware that the extent to which notes vary on a continuum from verbatim to paraphrase depends on how detailed the content is and their familiarity with the domain. Overall, Van Meter et al.'s findings suggest that these mature students have elaborate schemata for notetaking. Some slots in their schemata describe interactions among external task conditions and internal cognitive conditions.

It seems a logical necessity that metacognitive monitoring is engaged when a task is ambiguous. Because domain knowledge cannot be relied on to guide a definition of the task in this case, the student must infer features of the task before engaging with it. Making such inferences involves a meta-level model (standard). A study by Wong, Wong, and LeMare (1982) illustrates this hypothesis in the context of a reading comprehension task. Some students were given paragraphs to read and "think about what the story is about." Other students received a text including pre-paragraph questions and were told those questions illustrated kinds of questions they would be asked after the study period. Pre-paragraph questions provide standards for monitoring two products: comprehension of each paragraph and perceptions about the reading comprehension task, namely, that comprehension meant learning only a subset of the information rather than all of it. Not surprisingly, students who received this explicit guidance performed better on comprehension and recall tests than students who did not receive pre-questions. Students who studied texts without pre-questions expressed frustration. They complained that there "was too much too learn." Their meta-level model of reading led them to infer they should learn everything. That standard did not align with the task. Not only did they perform less well, they also experienced emotional distress.

Study tasks may be ambiguous because they are not familiar to students. We know of no research that explicitly links familiarity to metacognition in studying. There is indirect evidence, however, that students do engage metacognitively in forming perceptions about learning tasks. For example, Winne and Marx (1982) interviewed elementary school students about what they perceived their teacher wanted them to do after the teacher had issued an instructional cue in a lesson. Their descriptions indicated that students do judge the familiarity of brief instructional episodes and, on that basis, the students exercised metacognitive control over how they engaged in classroom activities. Another illustration is provided by Briggs' (1990) investigation of the questions students asked when assigned a simple letter-copying task using an unfamiliar word processing system. Only students with considerable experience in word processing asked questions, presumably products of metacognitive monitoring, that were relevant to the task. Briggs concluded that many students lacked appropriate standards against which to metacognitively monitor what the task was and, thus, were unable to frame useful questions about the task.

In instances where tasks are ambiguous, it seems safe to hypothesize students will set their own goals and build plans on that basis. As our

example of the grade school student suggested, students also may reframe goals that are explicit and clear before generating a plan for studying. Goal setting and plan building is our next topic.

Setting Goals and Devising Plans

Most models of learning in the contemporary literature characterize learning as goal directed. This is logically necessary when studying is metacognitively directed because metacognitive monitoring entails using meta-level models as standards (Paris & Byrnes, 1989). A spate of correlational studies link students' motivational orientations and self-reports of how they set goals and plans for studying. For example, Ainley (1993) classified high-school students' in terms of three motivational orientations: striving for mastery and understanding, enhancing ego by demonstrating ability, or just meeting task demands. Motivational orientation predicted students' reports of using deep approaches to studying in which they strove to select cognitive operations that linked new information to prior knowledge versus using surface approaches that committed to memory surface features of information as relatively discrete items (e.g., Biggs, 1991). Woulters, Yu, and Pintrich (1996) report stronger findings. Motivational orientation accounted for approximately 22% to 29% of the variance in 7th and 8th grade students' reports about studying tactics they used in English, mathematics, and social studies, and 18% to 20% of the variance in students' reports about metacognitively monitoring and controlling their studying tactics in these subjects.

Several lines of work suggest that having specific goals can be deleterious to productive metacognition, particularly when domain knowledge is weak and the learning task is complex (Kanfer & Ackerman, 1989; Sweller, 1989). The common explanation is that metacognition can be engaged "too early," straining cognitive capacity to the point where metacognition interferes with learning the domain's rules (Kanfer & Ackerman) or schemata (Sweller). Other research (e.g., Catrambone, 1995) reveals that, when studying complex procedures (using the Poisson distribution in word problems), students fare better when subgoals are explicitly labeled for them than when they must locate those subgoals on their own.

A study by Morgan (1985) suggests that, even if students do set goals when they study, they may not be very good at it. He provided brief training for undergraduate education students to set one of three forms of goals in a year-long educational psychology course: (a) setting proximal subgoals for each study session in "behavioral" terms and planning to record at the end of each study session which goals were met to criterion, (b) setting one

overall (distal) goal for each study session, and (c) setting a goal for the amount of time that would be spent at the beginning of each study session. All three groups and a control group, who studied “normally,” recorded how long they studied during each of three randomly but spaced weeks over the year.

Morgan checked whether students complied with training and observed four main findings. First and second, students who set specific subgoals for each session and monitored their achievement scored higher on a final examination and reported more intrinsic interest for the subject than did students in the other three groups, who did not differ from one another. Third, students who set goals to study for set times studied longer than other students. Finally, on marks in other courses where, presumably, all the students studied “normally,” there were no differences in achievement. In other words, students who profited by setting proximal subgoals for each study session in one course did not transfer the effect (and, presumably, the goal setting procedure) to those courses.

Not all goals are devised by students. Nolen and Haladyna (1990) distinguished between teacher goals and student goals. They suggested that, since teachers have the final say in student grades, students need to “figure out what the teacher wants” and study accordingly. Other research shows that when goals are set by others (teachers) and students adopt those goals, studying and resulting achievement improve if the goals target the process of learning rather than just producing acceptable outcomes (e.g., Schunk, 1996).

A study by Elbaum, Berg, and Dodd (1993) shows that the plans students build for studying are influenced by beliefs. They asked undergraduate students to allocate hours of instruction to either formal studying tasks that focus on the language per se (listening to and repeating sentences from an audiotape, and memorizing vocabulary and idioms, and having a native speaker correct errors in conversation) versus functional tasks that emphasize using the foreign language for communicative purposes (reading foreign language newspapers, keeping a diary written in the foreign language, and having conversations with native speakers who do not correct errors but encourage expression). The more students favored functional tasks, the less they perceived that knowledge of vocabulary and grammar were important goals, and the more they emphasized knowledge of sociolinguistic factors that shape language use and communication strategies that help new speakers compensate for linguistic limitations. If we speculate about causal relations, students whose beliefs lead them to set goals for learning about

sociolinguistic factors and communication strategies will opt to study in functional formats. Students whose beliefs lead them to set goals for gaining competence in vocabulary and grammar will prefer formal tasks as the way to study a new language.

We observe two notable gaps in the literature. First, we know little about how students form goals (Winne, 1997). A very general answer is provided by Carver and Scheier's (1990) model of self-regulated behavior. They postulate a 3-tier hierarchy of feedback loops, headed by an idealized concept of self, followed by general principles that govern behavior and, at the lowest level, behavior programs. When monitoring reveals that the profile of products at a level differs from the standard for that level, control is exercised by the next higher level. It resets standards at the lower level where the discrepancy was perceived. Our earlier example of the student who held an epistemological belief in quick and effortless learning matches this description for producing goals.

Second, research that illuminates when and how a student might engage in metacognition during the first two stages of studying where tasks are perceived and where goals and plans are set, is scant. We hypothesize that metacognitive activity in these stages will be proportional to two factors: (a) the degree to which COPES facets are unspecified, and (b) the student's familiarity with the task.

Enacting Study Tactics and Strategies

It seems logical that students must know about and have at least minimal skill in enacting study tactics and strategies for metacognitive control to enhance studying. Without tactics and strategies, there is little to control. The literature, however, is replete with experiments that demonstrate gains when students are trained to use a study tactic and compared to peers who study "normally." The consistency of findings that "normal" studying is relatively ineffective suggests that students may be very undereducated about study tactics and strategies (cf. Pressley, Van Etter, Yokoi, Freebern, & Van Meter, Chap. 14, this volume).

Some knowledge about tactics is, however, subtle and might not be expected to be apparent to students without direct instruction or supportive scaffolding (Winne, 1995a, 1995b, 1997). For example, Dempster (1988) argues that even educators are undereducated about the robust spacing effect, that is, that learning can be improved by distributing study sessions. Students who do not metacognitively monitor against a standard that describes spacing of study sessions may well learn less. They may also draw invalid conclusions about the utility of some tactics by misattributing weak

effects to otherwise effective tactics because they did not engage in spaced study sessions (e.g., Krug, Davis, & Glover, 1990).

A study by Spurlin, Dansereau, O'Donnell, and Brooks (1988) shows that students also need to understand how tactics affect learning. Summarizing has a solid history as an effective study tactic (see Foos, 1995) because, in broad strokes, students engage in varied forms of deeper processing of information as they construct summaries. For example, to select which information to include in a summary, the student must judge which ideas are central and which are less so. Making this determination can uncover instances where comprehension has failed, thereby inviting the student to make a repair for lack of comprehension. Another form of deeper processing can arise if summarizing occurs at the end of reading a text. In this case, effort spent in retrieving, selecting, and organizing information for the summary enhances retrieval paths and builds organizational structures that can support post-study retrieval based on partial cues (King, 1992).

Spurlin and colleagues compared students who generated four summaries, one for each fourth of a relatively large text (approximately 2,500 words), students who generated two summaries, one for each half of the text, and students who did not summarize. Results showed no differences in recall between students who generated four summaries and those who did not summarize at all, and both these groups recalled less than students who generated two summaries. Foos (1995) replicated this same pattern of findings, comparing one group who made two summaries of a 2,800 word text, one for each half, a second group who made one summary for the whole text, and a third group who made no summary. The common interpretation is that integrating all to-be-learned information in a single summary involves deeper processing and builds more useful retrieval cues than separately integrating information in several independent summaries. Hence, unless students use standards for judging whether tactics have been enacted appropriately, they may reach biased judgments about the effects of a tactic.

Research on experts' studying suggests that conditional knowledge that toggles tactics is probably *sub rosa* (McKoon & Ratcliff, 1992) in the sense that it is enfolded within an expert's domain knowledge rather than occupying separate status in long-term or working memory. However, Wyatt et al.'s (1993) study of domain experts (professors of social sciences) studying complex material outside their domain (a journal article on reading strategies) aligns with other research (e.g., Voss, Blais, Means, Greene, & Ahwesh, 1989) in demonstrating that experts deliberately monitor and control study tactics when domain expertise is challenged.

Some students metacognitively monitor the use of tactics using standards that refer to breadth of understanding and growth in knowledge. In contrast to these students who monitor with learning orientation, other students metacognitively measure studying in units that count the tasks they complete (Dweck & Leggett, 1988), a so-called performance orientation. A study by Tuckman (1996) shows that students can be guided by external conditions to favor one orientation over the other in metacognitively monitoring how they study.

Tuckman (1996) hypothesized that weekly quizzes administered at the start of every lecture would prompt students to study often, and that students trained to use two study tactics proven in experimental settings—identifying key terms and defining them, and elaborating each term with an example—would improve their comprehension of course material. To make each intervention “meaningful,” he based part of students’ grades on their quiz scores or scores awarded to glossary entries, respectively. In Experiment 2, students trained to use (and who were observed to use) the study tactics spent an average of 38 minutes more studying each week but had lower scores on the final exam compared to students who studied for frequent quizzes. Tuckman’s conversations with students revealed that those using the study tactics to generate elaborated glossaries adopted a performance goal to add items to glossaries rather than a learning goal of enhancing knowledge. Hence, those students’ metacognitive monitoring included a standard that was misaligned with elevating their achievement.

Finally, we note that it is not common for research on metacognition to investigate relatively global, personality-like dispositions that may infuse tactic’s conditional knowledge. However, several lines of research strongly implicate such variables as potent IFs included in metacognitive monitoring. Cacioppo, Petty, Feinstein, and Jarvis (1996) reviewed wide-ranging research concerning one such individual difference, need for cognition. This is a tendency for people to “seek, acquire, think about, and reflect back on information to make sense of stimuli ... [rather than] rely on others (e.g., celebrities and experts), cognitive heuristics, or social comparison processes” (p. 198). In their meta-analyses, people high in need for cognition, whom Cacioppo et al. call “cognizers,” substantially outperformed “cognitive misers” in recalling information they had studied (mean $d = 3.55$, $p < .001$), were more likely to assess arguments for substantive merits and engage in scrutinizing information (mean $d = .312$, $p < .010$), were more likely to be metacognitively aware of cognitive effort spent (mean $d = .272$, $p < .001$), and in general, were more likely to engage in metacognitive control to correct initial misperceptions or biases (though such attempts are not

necessarily successful). Need for cognition has not yet been examined in terms of traces obtained during studying, but it seems a good candidate for inclusion as a pervasive yet probably tacit dispositional IF that affects metacognitively guided studying.

Adapting Studying

As a result of metacognitive monitoring in Stage 3, students make changes to domain knowledge by adapting studying. Self-regulated learning, or metacognitive control of study tactics and strategies, is commonly viewed as occurring within the temporal boundaries of a study session. The methodological requirements for observing online, within-session self-regulation, however, are just now being conquered by advances in using computer technologies to trace how students study (Winne, Gupta, & Nesbit, 1994; Winne & Nesbit, 1995). Studies that use multiple-session designs, however, also demonstrate that students make forward reaching adaptations to studying tactics across study sessions as a result of metacognition about just-experienced (or prior) studying.

In one study, Rabinowitz, Freeman, and Cohen (1993) demonstrated an interaction between domain knowledge, first attempts to use studying tactics, and adapting approaches to study in a following study period. Students studied lists of randomly ordered words drawn from common categories (e.g., fruits, musical instruments). In a first session, students were told to use categorizing as a study tactic. One group studied items that were highly typical of the categories—domain knowledge was relevant here—whereas another group studied words low in typicality, making domain knowledge less useful. In a second study session, all students studied the same new list of medium typicality words using whatever study tactic(s) they preferred; no instructions were given to use categorizing as a study tactic. After recall of the new list, students were asked to describe study tactics they had used to study. And, after each of the two sessions, all students had rated how difficult, effortful, and useful it was to categorize words while studying.

The interesting finding concerns how students studied the second list. Recall of moderately typical words on the second list was the same for both groups, and it didn't differ from recall of the first list. Whatever study tactics students used in the second session were as effective as the categorizing tactic they were instructed to use studying the first list. Did students use a categorizing tactic on the second list? Among students who studied highly typical items in the first session, 96% reported they used categorizing only

or categorizing plus another tactic on the second list. Only one student abandoned categorizing outright. Among students who had studied less typical words in the first session, only 53% reused categorizing alone or supplemented it with another study tactic; the rest abandoned categorizing. Students' perceptions of whether they used categorizing as a study tactic were verified by clustering scores. Thus, students who first tried a study tactic that was ill-matched to the task, because domain knowledge of categories was not well-matched to categorizing the atypical words they were studying, were not likely to use that same tactic in a second task where it was more likely to be helpful because medium typicality words are more readily categorized.

In Rabinowitz et al.'s (1993) study, learners did not differ in the effort they said they applied to studying. Other research, however, suggests that effort is a condition that students metacognitively monitor and use in exercising metacognitive control. Eisenberger (1992) has reviewed research on the transfer of effort, characterizing such transfer as learned industriousness. In a typical study (Eisenberger, Masterson, & McDermitt, 1982), undergraduate women tackled problems involving addition, anagrams, and differences between pairs of cartoon pictures. One group of women experienced all three types of problems. Three other groups were asked only one kind of problem each. Within each of these four groups, half the students were given easy problems that required little effort (e.g., adding 2-digit numbers) and half were given difficult problems that required high effort (e.g., adding 7-digit numbers). In a second session, all the students were set a completely different task, writing an essay about a controversial topic. To gauge effort applied in writing essays, their compositions were scored for length, a direct measure of effort applied, and quality, an indicator of effort that probably reflects students' use of writing tactics and strategies as "the substance of effort, the means by which effort is applied" (Winne, 1995a, p. 176).

Among students who solved only one type of problem in the first session, regardless of whether problems required more or less effort, the effort they said they invested was not associated with length or quality of essays. Students who solved a variety of easy problems wrote the shortest and lowest quality essays, whereas those who solved varied and hard problems wrote the longest and highest quality essays. According to Eisenberger's concept of learned industriousness, students in the varied problems group who were posed difficult problems abstracted a general perception that effort was an inherent part of tasks, and that perception became a standard they used to metacognitively monitor work when they wrote essays.

These studies indicate that, across study sessions and across tasks, students exercise metacognitive control that changes conditions in Stage 1 of subsequent studying cycles or sessions. Given our hypothesis about the recursive properties of studying, these changes will percolate throughout that subsequent cycle or session.

SUMMARY

Our model depicts studying as metacognitively powered self-regulated learning that spans four recursively linked and weakly sequenced stages: defining the task, goal setting and planning, enacting study tactics and strategies, and metacognitively adapting studying for the future (see Table 12.1). Each stage of studying is modeled by a single cognitive architecture in which environmental factors and individual differences (conditions) constitute a temporary state of a task space. Students engage in activities (operations) that create information (products) that can be monitored in relation to standards (see Fig. 12.1). At each stage, students generate internal feedback and may be provided with external feedback (evaluations). This updates their understanding of a task, goals and plans they create, the study tactics they enact, and produces relatively permanent adaptations to cognitive structures that describe tasks and studying. When models of tasks and studying are used as the basis for examining information at each stage, students engage in metacognitive monitoring. When students toggle tactics on and off or when they edit tactics, this metacognitive control effects self-regulating learning. We examined research that illustrates metacognition in each stage.

FUTURE RESEARCH ON STUDYING AS SELF-REGULATED LEARNING

Research demonstrates individual differences in students' perceptions of a task, and we noted that a teacher's (or researcher's) perception of an assigned task may not align with students' understandings of "the" task. These differences originate with variance in students' beliefs, motivation, and prior knowledge, all of which, according to our model, are products of prior engagement with previous study tasks. Though we speculated about metacognition in Stage 1 of studying, we do not find that research has directly investigated how students exercise metacognitive monitoring and control when they generate an understanding about a study task.

It is frequently assumed that students' perceptions of goals (standards) for a study task match assigned goals. At least, when there is no check on this correspondence, it seems this assumption is made. There may be adequate equivalence between goals assigned and students' perceptions; however, our model allows that, at Stage 2, students can add to or modify goals that they may accurately acknowledge as having been set for them. Although there are studies that document disparities between assigned goals and adopted goals, we know of none that explicitly investigates roles played by metacognitive monitoring and control in goal setting and subsequent planning.

Metacognition is a driver for online changes that students make to tactics they use to study. This stage of studying has more frequently been a focus for research, and there is considerable evidence that metacognition is a powerful influence here. However, the field needs to know more about kinds of meta-level knowledge that students use in metacognitive monitoring and about students' editing study tactics.

Finally, the field is struggling with issues of how students maintain, transfer, and adapt their approaches to studying. These are questions about Stage 4 of our model. The backward- and forward-reaching nature of metacognitive monitoring and control in this larger scale adaptive work is clearly more complicated than the local adaptations addressed in Stage 3, where students enact and control tactics. Perhaps progress in understanding these larger scale adaptations requires developing more adequate views about metaknowledge *per se*. As clarified by Alexander, Schallert, and Hare (1991) in their review of terms researchers use to describe knowledge, this is a formidable task.

We have three major recommendations for future research about any stage or stages of studying. First, traces (Winne, 1982) of metacognitive monitoring and control should be a standard feature. Without such data, we have only indirect evidence about metacognitive events and their recursive and forward reaching features. Trace data can chart how students metacognitively monitor and regulate within and between stages of studying. In addition, because traces operationally define metacognition, they provide clear guidelines for framing objectives about metacognition and for assessing those achievements in study skills training and other instructional interventions.

Second, studies should be designed with explicit options for students to toggle and edit study tactics. In many experiments, the control group meets our criterion of allowing students to exercise toggling. They are expressly told to study as they wish. In contrast, students who experience or take part

in an intervention, such as training to use concept maps for organizing information during a study period, implicitly have toggling and editing curtailed. The experimenter does not want students to substitute outlining or point-form notetaking for concept mapping, although the process of examining whether to make such a substitution and how that decision-making process itself may be adapted is the substance of metacognition. In this context, trace data will be essential in revealing whether and how students toggle and edit.

Finally, studies should generate data that allow direct inspection of students' perceptions of conditions, operations, products, evaluations, and standards. These are the inputs to metacognitive monitoring and the products that are updated by metacognitive control. Our model suggests two implications. First, prior knowledge is not a static condition. It is potentially updated throughout a task. Second, though we labeled broad categories of "prior knowledge," it remains a challenge to understand how students "collapse" or "integrate" the information in these multivariate profiles to effect self-regulation.

ACKNOWLEDGMENT

Support for this research was provided by a grant to the first author from the Social Sciences and Humanities Research Council of Canada (#410-95-1046).

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